

A Visual Study of E-Commerce Orders, Delivery & Customer Satisfaction

Technical Report

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Abstract

Olist's Brazilian e-commerce marketplace connects thousands of small and medium sellers to customers, and its leadership faces a persistent tension: growing the marketplace aggressively risks the customer experience, while enforcing strict delivery quality risks stalling growth. This report analyzes 99,441 orders (September 2016 to October 2018) across the full order journey placement, payment, seller handling, freight, delivery, and review to identify what drives customer satisfaction and where the marketplace is quietly at risk. We find that delivery delay predicts review score almost monotonically, but that the mechanism is more specific than it first appears: transport time, mostly outside a seller's control, drives the damage to any individual order's rating, while a seller's own processing speed is the better signal for identifying which sellers are reliably good over time. We further show that delivery quality did not hold steady as the marketplace scaled, while customer acquisition efficiency improved over the same period which is a measurable, not hypothetical, growth versus quality tension. Beyond these central findings, we report several smaller, honestly-caveated discoveries, including a counter intuitive absence of holiday season leniency, an earl delivery ceiling effect, and a seller service compliance metric distinct from delivery outcome that surface patterns not previously documented in this dataset. We translate these findings into concrete investment and intervention recommendations for leadership, and deliver an interactive dashboard that lets a non technical viewer explore the same findings by category, region, seller, and delivery performance.

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1 Introduction

1.1 Business Context

Olist operates an online marketplace that connects thousands of small and medium-sized sellers across Brazil to customers, handling everything from product listing to payment to logistics through a single unified platform. Leadership's stated goal is straightforward to state and difficult to execute: **grow the marketplace without breaking the customer experience.**

In practice, every lever available to leadership trades one objective against the other. Onboarding sellers aggressively raises gross sales, but late deliveries, poorly rated products, and negative reviews begin driving customers away. Conversely, enforcing strict quality standards on every seller improves the customer experience, but shrinks the catalogue and stalls growth. At present, these trade offs are navigated on intuition rather than evidence.

1.2 Problem Statement

This project's task is to study the complete journey of an order from placement, through payment and seller handling, through freight and delivery, to the review a customer ultimately leaves and turn that journey into a clear, visual account of two linked questions:

1. What actually drives a customer from a satisfied order to a one star review?
2. Where in the marketplace is risk quietly building up, and where is there room to grow safely?

The ultimate goal is to give leadership a concrete basis for deciding where to invest and where to intervene, replacing gut feel with evidence.

1.3 Objectives

- Understand the dataset's structure and translate the business goal into specific, answerable analysis questions.
- Clean and prepare the data, resolving known quality issues rather than working around them silently.
- Build exploratory visuals to uncover structure, and explanatory visuals that directly answer the analysis questions.
- Go beyond descriptive, industry-standard findings to surface smaller, less obvious patterns worth further investigation while being explicit about their uncertainty.
- Deliver an interactive dashboard, this technical report, and a final presentation that together let leadership act on the findings.

2 Data Preparation

2.1 Dataset Overview

The analysis draws on the Olist Brazilian E-Commerce dataset together with a companion Marketing Funnel dataset eleven raw CSV files in total, covering orders, order items, payments, products, sellers, customers, geolocation, reviews, marketing-qualified leads, and closed deals. The core order data spans **99,441 orders** placed between September 2016 and October 2018.

Work was divided into five tracks following the natural order journey, each producing a cleaned, shared output file:

Category	Focus	Key output file(s)
A	Payments & Orders	<code>orders_master.csv</code> (99,441 rows)
B	Products & Categories	<code>order_products_summary.csv</code> , <code>category_lookup.csv</code>
C	Sellers & Logistics	<code>sellers_master_enhanced.csv</code> , <code>order_logistics_master.csv</code>
D	Customers & Reviews	<code>customers_master.csv</code> , <code>reviews_clean.csv</code>
E	Marketing Funnel & Integration	<code>funnel_summary.csv</code> ; builds the final joined table

Table 1: The five analysis tracks and their outputs.

These were joined into a single analytical table, `analysis_master.csv` one row per order, **99,441 rows × 69 columns**. Every summary statistic in this table (review score distribution, overall late rate, the count of delivered but unreviewed orders) was independently re derived from the merged file and confirmed to reproduce each track’s originally reported figures exactly, verifying the join introduced no silent data loss or duplication.

2.2 Data Cleaning and Quality Challenges

Several data-quality issues were identified and resolved with documented reasoning rather than silently dropped or ignored:

- **Raw CSVs must never be opened in Excel.** Doing so silently reformats date columns to day first order and truncates timestamp precision to the minute, corrupting the file if re saved. This was discovered early and flagged across every track.
- **The review- gap selection bias was resolved, not just documented.** 646 delivered orders have no review. A two-step test confirming review requests fire on a near-fixed, near-immediate schedule after delivery (median 0.24 days), and confirming the 646 no-review orders are not concentrated near the dataset’s extraction cutoff rules out “insufficient time to respond” as an explanation. The gap is genuine non-response, concentrated where satisfaction was worst.
- **Known anomalies were catalogued with exact counts:** two chronologically impossible orders (a carrier handoff timestamp predating the purchase timestamp by more than a day), one order with no payment record at all, and 1,174 delivered orders with a mild (sub two day) carrier before approval logging quirk, most plausibly a benign timestamp-ordering artifact rather than a real process failure.
- **A dataset extraction artifact was identified and excluded from time series analysis.** The final two months of order data (September to October 2018) show 94 to 100% cancellation rates not a business collapse, but orders too close to the data pull date to have completed their natural lifecycle. Two additional single digit order months at the very start of the dataset (September and December 2016) were excluded from monthly trend charts for the same small sample reasoning.

2.3 Feature Engineering

Several derived features underpin the analysis:

- **Delay buckets:** `delivery_delay_days` (delivered minus estimated date) is binned into five bands from “5+ days early” to “8+ days late,” the primary lens for the central finding.

- **Processing vs. transport split:** `processing_days` (purchase to carrier handoff, seller-controlled) and `transport_days` (carrier handoff to delivery, carrier-controlled), the decomposition behind this project's central mechanism.
- **Freight ratio:** freight cost divided by item price, a size independent measure of shipping burden.
- **Seller confidence flag:** sellers with fewer than five total orders are flagged `low_confidence` (42% of all sellers) and treated with explicit caution in every seller-level rate or average throughout the project, rather than silently included.
- **Repeat customer flag** and **primary category** (the category of the highest-priced item in a multi-item order, a documented, non-arbitrary convention).

3 Methodology

3.1 Analytical Approach

Each track was built in two passes: an initial cleaning and exploration pass, followed by a deeper follow-up pass that dug further into the track's own data and, in several cases, joined against other tracks' outputs for the first time. Two disciplines were applied consistently:

- **Cross track validation.** Where two independently-built tracks converged on the same finding without having coordinated on it most notably the March 2018 anomaly, found separately in delivery timing data (Category A) and in review-score data (Category D) that convergence is treated as strong evidence the finding is genuine rather than a processing artifact in either track.
- **Verify before reporting.** Several apparent findings during the deeper analysis turned out to be some kind of mistakes (a scaling error, an incomplete file load) rather than real patterns, caught only by cross checking against an already known baseline. This practice was continued through the visualization phase: all the charts in each categories were built from scratch, executed end to end, and checked for zero errors before being finalized.

3.2 Visualization Design Principles Applied

Chart choices throughout this project were governed by course frameworks rather than default habit:

- **Task visualization fit:** the required viewer action (precise comparison, spotting an outlier, following a route) determined the chart type, not aesthetic preference.
- **Cleveland McGill encoding accuracy:** position and length were preferred over color and area wherever a chart's purpose was precise magnitude comparison; color was reserved for categorical distinctions.
- **Small sample discipline/Stevens:** every rate or average is shown alongside its sample size; segments below a working threshold are flagged rather than presented with false confidence, both in the static notebooks and, more importantly, in the interactive dashboard where a viewer can filter into arbitrarily small segments.
- **Tufte's heuristics:** data-ink was maximized where it aided interpretation (direct value labels rather than requiring axis-reading) and minimized where it did not (unnecessary gridlines, decorative styling); small multiples were used in place of 3D or overly dense single charts.

4 Exploratory Data Analysis

This section summarizes the headline exploratory finding from each of the five tracks. Full detail lives in the accompanying notebooks (see Appendix C).

4.1 Delivery Performance

The order funnel is clean: 97% of orders reach delivery, with a small, well-understood tail of cancellations and unavailable items. Olist’s promised delivery dates carry a large built-in safety buffer the median delivered order arrives **11.95 days before** its estimated date, and only **8.11%** of delivered orders are actually late. This means “on time” is the default outcome, not an achievement, and a late order is a comparatively rare, meaningful signal.

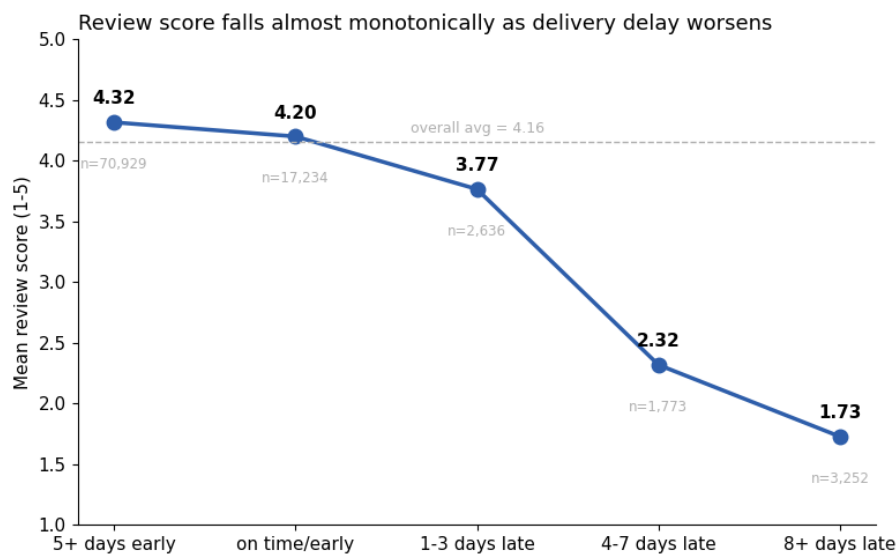


Figure 1: Mean review score by delivery-delay bucket the project’s anchor relationship.

4.2 Geography & Sellers

Cross-state deliveries take nearly twice as long on average as same-state deliveries (15.1 vs. 7.9 days) and are correspondingly less reliable (9.3% vs. 6.1% late). This average, however, conceals dramatic route-level variation: specific seller-state-to-customer-state routes run at 30 to 55% late rates, several times the marketplace baseline. Marketplace performance is also highly concentrated: the top 10% of sellers by volume account for approximately **68%** of all orders.

4.3 Category & Economics

Of 74 product categories, volume is heavily concentrated in a top slice; cancellation rate varies modestly (0.19%–0.80%) across the fifteen largest categories. Three categories `market_place`, `toys`, and `consoles_games` are declining in absolute monthly volume even as the marketplace overall grows, a genuine “growth isn’t reaching everywhere” signal distinct from simply growing more slowly than average.

4.4 Customer Behavior

Review scores are strongly bimodal, not centered: 57.8% five-star, 11.5% one-star, with a thin middle. The presence of a written comment is itself a strong dissatisfaction signal (3.70 vs. 4.38

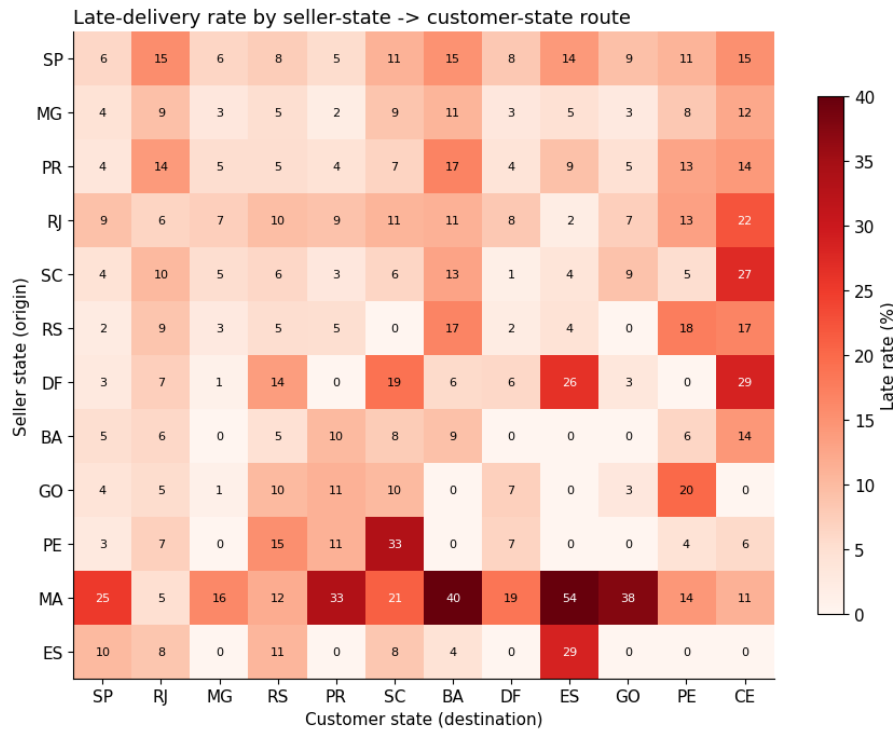


Figure 2: Late-delivery rate by seller-state to customer-state route.

average score with vs. without a comment). Only 6.3% of customers ever place a second order, making repeat purchase a genuinely rare outcome worth understanding on its own terms.

4.5 Acquisition Funnel

Overall lead-to-seller conversion is 10.5%, but varies sharply by channel: paid and organic search convert well above average (12.3% and 11.8%), while email and other minor channels convert well below (3.0% and 2.7%) despite carrying real volume. Only 12.3% of currently active sellers have any funnel record at all, meaning funnel-derived findings apply specifically to that overlap population, not the marketplace as a whole.

5 Explanatory Analysis: Deep-Dive Insights

5.1 The Central Mechanism and Its Paradox

The project's anchor finding is that review score falls almost monotonically as delivery delay worsens, from 4.32 (5+ days early) to 1.73 (8+ days late). Decomposing delay into its seller-controlled leg (processing time) and carrier-controlled leg (transport time) reveals a more specific and more actionable mechanism: **transport time is the single strongest order-level correlate of review score ($r = -0.299$)**, ahead of total delay itself ($r = -0.267$) and well ahead of processing time ($r = -0.157$).

At the *seller* level, however, this ranking flips. Restricting to sellers with at least five orders, **processing time is the stronger correlate of a seller's average review score ($r = -0.449$)**, compared to transport time ($r = -0.296$).

Practical implication: transport time, largely outside any individual seller's control, drives the customer-facing damage on any single order — so disciplining sellers for overall lateness partly targets the wrong lever for fixing the customer experience.

Processing time, being within a seller's own control, remains a legitimate and more consistent signal for screening which sellers are reliably trustworthy over time.

The flip: transport dominates per order, processing dominates per seller

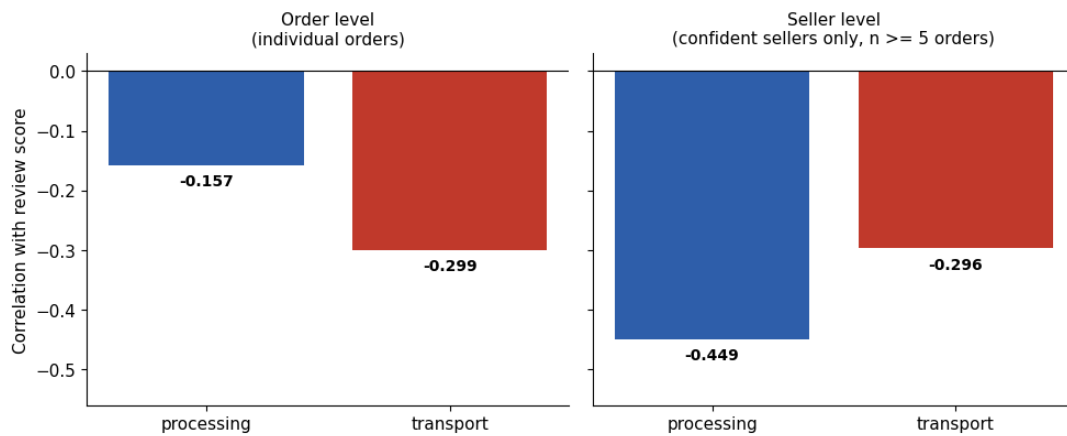


Figure 3: The correlation ranking flips between the order level and the seller level.

5.2 Growth vs. Quality: A Measurable Tension

Delivery quality did not hold steady as the marketplace scaled. Two real spikes interrupt an otherwise steady growth in monthly order volume: November 2017 (13.8% late, plausibly coinciding with Black Friday demand) and March 2018 (20.7% late, cause unknown from this data alone). The March 2018 anomaly is independently corroborated: the same month shows the lowest average review score of any regular month (3.81) in data built entirely separately from the delivery-timing analysis two independently-constructed tracks landing on the same real-world event.

On the acquisition side, the opposite pattern holds: monthly lead-to-seller conversion efficiency rose from under 1% in mid-2017 to 13–14% by early 2018, improving *as* volume scaled rather than degrading. Together, these two findings make the growth-versus-quality tension concrete and measurable rather than merely asserted.

5.3 Additional Verified Findings

Three further findings, built from raw files beyond the core `analysis_master.csv` table, add texture to the picture above:

- **Seller compliance.** Beyond whether an order arrived late, `order_items_dataset.csv` records each item's promised `shipping_limit_date`. 9.35% of items ship after the seller's own committed deadline. This metric correlates with the existing late-rate measure ($r = 0.34$) but is meaningfully distinct a seller can meet delivery outcomes while still consistently missing their own commitments, or vice versa.
- **Freight cost is driven by physical size, not arbitrary pricing.** Joining `order_items_dataset.csv` to `products_dataset.csv`, product weight correlates with freight cost at 0.61 and volume at 0.59 confirming that categories with unusually high freight ratios are burdened by genuine physical bulk, not inconsistent charging.
- **Comment length scales with dissatisfaction.** Beyond simply having a comment, the length of that comment falls sharply and monotonically with review score 1-star

comments average 76.5 characters versus 18.6 for 5-star cheap additional triage signal requiring no text/sentiment processing.

5.4 Discovery-Pillar Findings

A dedicated pass specifically hunted for patterns not already standard knowledge in e-commerce logistics. The following four findings are reported with their caveats intact, as directional evidence rather than settled conclusions:

1. **No holiday tolerance effect.** At every delay bucket, the Nov 2017–Jan 2018 period scores *slightly lower*, not higher, than the rest of the year the opposite of the common assumption that customers are more forgiving during a known high-volume season.
2. **An early-delivery ceiling effect.** Review score rises with earliness up to roughly 15–21 days early (peak 4.34), then mildly reverses through 22–45 days early (down to 4.25). “Earlier is always better” is not quite true.
3. **Category delay-sensitivity tracks semantic urgency.** Impulse and occasion-driven categories (`toys`, `sports_leisure`, `health_beauty`) punish delay more heavily than considered, bulky-goods categories (`furniture_decor`, `housewares`) a plausible but unconfirmed read of the correlation.
4. **Loyalty buys patience for large failures, not small slips.** Repeat customers score *worse* than first-time customers at small delays (1 to 3 days late: 3.62 vs. 3.78) but *better* at large ones (8+ days late: 1.90 vs. 1.72) a more nuanced story than “repeat customers are simply more forgiving.”

6 Business Impact & Recommendations

6.1 Where to Invest

- **Logistics and carrier-network improvements**, particularly for the specific high-risk routes identified in Section 4.2, since transport time not seller handling drives most of the customer-facing lateness problem (Section 5.1).
- **Tooling and account support for the top 10% of sellers** by volume, given their outsized influence on overall marketplace performance.

6.2 Where to Intervene

- Screen seller reliability using **processing time**, not overall lateness, since the latter is heavily confounded by transport factors outside a seller’s control.
- Consider **category-specific delivery SLAs**, informed by the uneven delay-sensitivity documented in Section 5.4.
- Investigate the specific worst-performing routes and sellers named in the accompanying notebooks as targeted intervention candidates, rather than applying uniform marketplace-wide policy.

6.3 What to Monitor

- Declared seller attributes at signup (`lead_type`, `business_type`) as an early-warning flag for new sellers, before sufficient order history accumulates to assess them directly.
- The unexplained March 2018 dip, in case a recurring pattern emerges as more data becomes available.

7 Interactive Dashboard

A six-page interactive dashboard (built with Streamlit) accompanies this report, mirroring its analytical structure: Delivery Performance, Geography & Sellers, Category & Economics, Customer Behavior, and Acquisition Funnel, alongside a landing overview page. It allows a non-technical viewer to filter by product category, customer state, seller state, seller size, and delivery-performance bucket, with every filter shared live across all pages via URL-based state, so a filtered view can be bookmarked or shared directly.

Every rate or average displayed states its underlying sample size, and any segment falling below a safe threshold is explicitly flagged rather than shown with unwarranted confidence a direct extension of the small-sample discipline applied throughout the static analysis into a setting where a viewer, not the analyst, controls which slice of the data is being examined.

8 Limitations & Future Work

- **March 2018’s cause remains unknown.** The spike is real and cross-validated from two independent angles, but no column in this dataset explains it. A genuine limitation, stated plainly rather than replaced with a guess dressed up as an insight.
- **Funnel findings generalize only to a 12.3% overlap** between funnel-recorded leads and currently active sellers; they should not be read as marketplace-wide.
- **The discovery-pillar findings (Section 5.4) are directional, not definitive,** resting on smaller samples and plausible-but-unconfirmed mechanisms.
- **Review comment text remains largely unprocessed** beyond simple length — a natural extension for future work, ideally with Portuguese-language sentiment or topic modeling.
- **Future data that would resolve open questions:** an external event log for March 2018 (e.g. a logistics-partner incident or regional disruption), landing-page-level funnel detail, and richer product-description data to test whether more informative listings reduce mismatched-expectation dissatisfaction.

9 Conclusion

This analysis traced the full Olist order journey across five coordinated tracks and re-verified every headline number against the underlying raw data before reporting it. The central finding that transport time, not seller processing time, drives the customer-facing cost of delay, while processing time better identifies which sellers are trustworthy gives leadership a specific, actionable lever rather than a generic “fix delivery” directive. The growth-versus-quality tension is shown to be real and measurable, not hypothetical, with evidence on both sides from independently-built analyses. Beyond these central results, several smaller, honestly-caveated discoveries surface patterns not previously documented for this marketplace, fulfilling the goal of finding something genuinely new rather than only confirming industry-standard intuitions. The accompanying dashboard, notebooks, and code repository make every figure in this report directly reproducible.

A Appendix - Team Contributions

As required, individual contributions are documented below. Each team member additionally maintained a work log (date, task, output, time spent) in the shared tracking document referenced in Appendix C.

Name	Category(s) owned
Smit Pareshbhai Sabalpara	e.g. Category A – Payments & Orders
Harsh Patel	e.g. Category B – Products & Categories
Renee Keerthana Paturi	e.g. Category C – Sellers & Logistics
Naman Goel	e.g. Category D – Customers & Reviews
Sunny Kumar Gupta	e.g. Category E – Marketing Funnel

B Appendix - Data Dictionary Summary

The full column-by-column data dictionary for every processed file, including `analysis_master.csv`'s complete 69-column schema and documented null patterns, is provided in the repository (see link below). Key files:

- `analysis_master.csv` 99,441 rows $\times$ 69 columns, one row per order, the primary analytical table.
- Five track-level processed files (Section 2.1) intermediate outputs, each independently verified.
- Four raw files used beyond `analysis_master.csv`, each needed where a specific finding required a column not present in the joined table:
 - `order_items_dataset.csv`
 - `products_dataset.csv`
 - `order_reviews_dataset.csv`
 - `marketing_qualified_leads_dataset.csv` and `closed_deals_dataset.csv`

C Appendix - Links & Artefacts

All hyperlinks below have been verified accessible to evaluators as of the submission date.

- **GitHub repository (code, notebooks, dashboard source):** https://github.com/22f3001607/DVD-Project_Group-5
- **Raw and processed datasets (Google Drive):** <https://drive.google.com/drive/folders/1ALV7ElJbM7NVjogPuGVL0c40f8zfSrrZ>
- **Final presentation slides:** <https://docs.google.com/presentation/d/1Em013hh2hkSxfIne9P6sM8VRVBi1Y4eC4vSuiQJtBsk/edit?usp=sharing>
- **Work log - Smit Pareshbhai Sabalpara:** https://docs.google.com/document/d/1BwuGBixZJGJ2LQ_K_7y87ZxeZTuwjWVycdagDvlt18Y/edit?usp=sharing
- **Work log - Harsh Patel:** https://docs.google.com/document/d/1272XoEhsXaq5NE_k6sQD9x4KsjBpu-Ca/edit?usp=drivesdk&oid=109738766625376228025&rtpof=true&sd=true

- **Work log - Rennee Keerthana Paturi:** <https://docs.google.com/document/d/1ZXGadKAFUc2JrtMiJNuVmmps2eSD79PdZ/edit>
- **Work log - Naman Goel:** <https://docs.google.com/document/d/1iFlX8CGFGijqcqFoY-5WhzH0M/edit?usp=sharing>
- **Work log - Sunny Kumar Gupta:** <https://docs.google.com/document/d/1cMXt9itJB4oFTYkzFeLRQn9oYWGS0JPQfTQKmrYdPuU/edit?usp=sharing>
- **MoMs:** <https://docs.google.com/document/d/1jqHK8HFp0JF11BqviuB9dXy72NPOGqrwZVExxvZ9F/edit?usp=sharing>